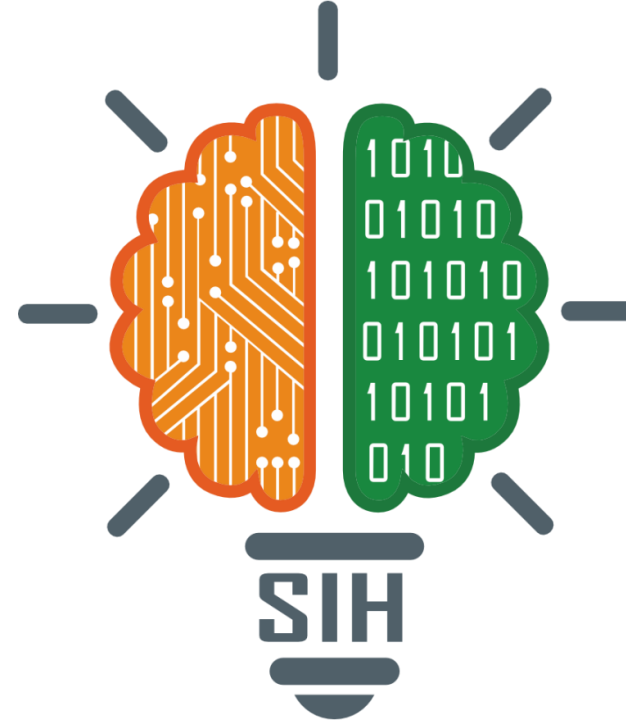
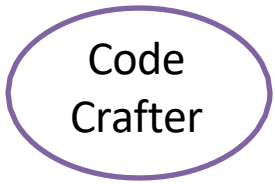


TITLE PAGE

- **Problem Statement ID** – 25040
- **Problem Statement Title**- FloatChat - AI-Powered
Conversational Interface for ARGO Ocean Data
Discovery and Visualization
- **Theme**- Miscellaneous
- **PS Category**- Software
- **Team ID** - IPEC/2025/115
- **Team Name**- Code Crafter





IDEA TITLE

IDEA TITLE & DESCRIPTION

Idea Title: FloatChat

Proposed Solution: An AI-driven chatbot for querying ARGO oceanographic datasets (temperature, salinity, currents, etc.) via natural language. Users ask questions and receive answers plus interactive visualizations, without needing expertise in data formats or coding.

Problem Addressed: Simplifies and democratises access to ARGO's complex, large-scale ocean data. By enabling plain-language queries and real-time visual feedback, FloatChat lowers the barrier for non-experts to explore marine data. This approach is similar in spirit to NASA's Earth Copilot, which uses AI to make Earth science data accessible to a broad audience.

Innovation: Provides a **real-time conversational search** and dynamic visualization interface tailored to ARGO data. Unlike static dashboards, FloatChat uses advanced NLP (LLMs) to interpret queries and generate custom plots or maps on-the-fly. This makes ocean data exploration intuitive and educational for researchers, students, journalists, and policymakers.

TECHNICAL APPROACH



CODE
CRAFTER

•**Core Technologies Backend**: Python, FastAPI, PostgreSQL AI/ML: TensorFlow for advanced models Frontend: React for intuitive UIs Cloud-Native: Docker & Kubernetes for scalability



•**Intelligent Models Anomaly Detection**: Identifying critical deviations in ocean parameters. Trend Prediction: Forecasting oceanic changes for proactive measures.



- **Leveraging Open Datasets**

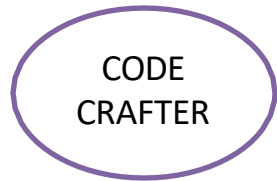
- Our system integrates with globally recognized open data sources:
 - **Argo:** Global array of profiling floats.
 - **Copernicus:** EU Earth Observation Programmes.
 - **NOAA:** National Oceanic and Atmospheric Administration data.

- **Cost-effective, Cloud-Based, and Globally Scalable**

- Designed for efficiency and global reach, our solution minimizes operational overheads

- **Addressing Challenges**

- **Risks:** Handling missing data, real-time streaming complexities.
- **Mitigation:** Hybrid storage solutions, AI-driven interpolation for data gaps, scalable APIs for robust performance.



IMPACT AND BENEFITS



Climate Scientists

Enhanced data access and predictive tools for critical research.



Policymakers

Informed decision-making for environmental regulations and climate policy.



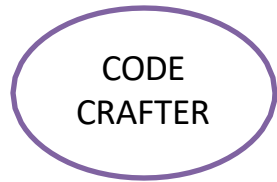
Maritime Agencies

Improved safety and operational efficiency for shipping and fisheries

Social Impact: Bolstering disaster preparedness for cyclones and tsunamis through predictive insights.

Economic Impact: Supporting vital industries including shipping, fisheries, and coastal tourism.

Environmental Impact: Strengthening global climate change monitoring and research efforts. Scalable to include rivers, lakes, and groundwater monitoring.



RESEARCH AND REFERENCES



THE FUTURE IS OCEANIC: JOIN CODE CRAFTERS

We are Code Crafters, and we are building the future of ocean intelligence.

- Unifying disparate ocean data streams.
- Leveraging cutting-edge AI for predictive insights.
- Delivering tangible impact for climate research and safety.
- Scalable, cost-effective, and poised for global reach.

Thank you.